Renewable and Sustainable Energy Reviews 225 (2026) 116159

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: www.elsevier.com/locate/rser

# A systematic review of electricity demand for large language models: evaluations, challenges, and solutions

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ARTICLE INFO

*Keywords:*
Electricity consumption
Artificial intelligence
Large language model
Power grid
Energy sustainability

ABSTRACT

Large language models (LLMs) are revolutionizing technology landscapes, transforming production workflows, and deeply embedding themselves into our daily lives. Their unparalleled capabilities, fueled by extensive training on vast datasets and intensive utilizations, underpin this transformation. However, this achievement comes at a significant cost: the immense electricity demand required for their training and inference poses an urgent and critical challenge. As the reliance on LLMs grows, ensuring a reliable and sustainable power supply becomes paramount for their uninterrupted progress and widespread adoption. This comprehensive review paper thoroughly examines the intricacies of LLMs' lifecycle, focusing on both training and inference stages. It critically assesses various parameters and approaches for accurately estimating their electricity consumption and associated carbon emissions, drawing upon representative statistical data spanning diverse LLM products and task types. By delving into the complexities, the paper uncovers the fundamental challenges in forecasting and fulfilling LLMs' extensive electricity demand, which are intricately linked to controversies within four bottom-up tiers: soft-and-hardware, data center, power grid, and external societal factors. Crucially, this paper offers a suite of tailored solutions for each tier, aimed at not only addressing the immediate electricity demand challenge but also fostering the long-term sustainability of LLMs. The insights outlined herein endeavor to alleviate the power shortage crisis facing LLMs, paving the way for their widespread adoption while minimizing their environmental impact and contributing to a greener future.

List of abbreviations

| | |
|---|---|
| AC | alternating current |
| AI | artificial intelligence |
| AWS | Amazon web services |
| CFE | carbon-free electricity |
| $CO_2$eq | $CO_2$ equivalent emissions |
| CPU | central processing units |
| DC | direct current |
| FLOPs | floating point operations |
| FLOPS | floating point operations per second |
| FPGA | field programmable gate arrays |
| GPU | graphic processing unit |
| GRPO | group relative policy optimization |
| LLM | large language model |
| MLA | multi-head latent attention |
| MoE | mixture-of-experts |
| NPU | neural processing unit |
| PUE | power usage effectiveness |
| RES | renewable energy source |
| SLM | small language model |
| TPU | tensor processing unit |
| UPS | uninterrupted power supply |

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## 1. Introduction

Large language models (LLMs) have rapidly swept across the globe, vastly expanding the frontiers of generative artificial intelligence (AI) [1]. The *largeness* of LLMs manifests in three dimensions, i.e., the immense scale of model parameters, the abundance of data, and the formidable computational demands. Since the introduction of Google's BERT with 0.1 billion parameters in 2018, the volume of LLMs has

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https://doi.org/10.1016/j.rser.2025.116159
Received 12 September 2024; Received in revised form 13 July 2025; Accepted 1 August 2025
Available online 11 August 2025
1364-0321/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

soared in just a few years. For instance, the scale of parameters for GPT-4 has increased to an astonishing 1.75 trillion, which consumes approximately 13 trillion tokens during its training, necessitating the utilization of 3125 V100 graphic processing units (GPUs) [2,3]. These colossal LLMs have not only unlocked unprecedented capabilities for intelligent computing, but their innovative achievements are also accompanied by a notable surge in power consumption, posing challenges to their electricity supply.

With the rapid emergence and widespread application of various LLMs, their electricity demand is growing at an increasingly rapid pace. The training of GPT-3 consumed approximately 1287 MWh, accompanied by over 552 tons of carbon emissions [4]. The even larger GPT-4 requires more than 40 times electricity of GPT-3. As for the deployment of LLMs, a single search on ChatGPT consumes about 25 times more energy than a traditional Google search [5]. If every Google search were to employ generative AI, the annual electricity usage would surge to about 29.2 TWh, equivalent to Ireland's total annual electricity consumption [6].

In the realm of LLMs, electricity has emerged as a pivotal restrict. The primary challenge stems from their immense power consumption [7]. Currently, the power of LLM-serving data centers has risen to hundreds of megawatts and continues to escalate, directly leading to acute shortages in their local power networks [8]. Many existing and new data centers necessitate the expansion and upgrade of their power grid infrastructure, including power distribution units, substations, and power generators [9].

Furthermore, the surge in electricity consumption exacerbates environmental burdens. Data centers currently account for approximately 1.4 % of global carbon emissions, with roughly a quarter of their workloads intimately tied to various AI applications [10]. The rise of LLMs is continuously elevating this proportion. If the electricity does not originate from clean renewable energy sources (RESs), it will give rise to even more pronounced carbon emission issues, exerting non-negligible impacts on the sustainable environment.

Some studies have endeavored to quantify the current and future electricity demand and associated carbon emissions of LLMs. The electricity consumed by LLMs can be primarily categorized into two stages, i.e., training and inference. Research on the training stage is relatively abundant due to its scheduled nature at specific locations and times, coupled with the staggering amount of electricity required for a single training run. On the other hand, while individual inference tasks consume significantly less power than training, their high frequency leads to a cumulative effect that contributes substantially to the overall electricity usage of LLMs [11]. Consequently, separate investigations into the training and inference stages are necessary when estimating LLM electricity demand [12].

Despite the consensus among researchers and engineers regarding the substantial electricity of LLMs, there exists considerable discrepancies in the specific evaluations of demand and carbon emissions [13]. For instance, a study in 2019 calculated that the training of BERT required 1.2 kW of power, resulting in 1507 kWh over 79 h and equivalent to approximately 652 kg $CO_2$eq [14]. However, a study in 2021 contended that the actual emissions were 88 times lower than this earlier estimation [15]. Given this significant divergence in assessing the electricity consumption of existing LLMs, estimating long-term energy demands becomes even more challenging [16,17].

By comparatively analyzing the specific evaluation factors from various literatures, causes of discrepancies in evaluating electricity demand for LLMs can be delved. The capabilities of LLMs are influenced by a combination of factors, e.g., model choices, data scales, hardware deployment [18,19]. For the same LLM, when trained or deployed in different data centers, the impact of geographical locations leads to changes in factors such as variations in power grid capacity and fluctuations in the proportion of clean energy sources, which affect the availability and sustainability of electricity to a large extent [10]. Additionally, the widespread adoption of LLMs, as a cross-industry and societal phenomenon, not only drives the growth of electricity demand but also faces external challenges stemming from technological advancements and substitutions.

To address the barriers in the power supply of LLMs and facilitate a transition towards low-carbon electricity, a multifaceted strategy portfolio is essential. However, current literature primarily focuses on energy estimation and reduction strategies at the data center level, e.g., optimizing power usage effectiveness (PUE), balancing computational loads, and purchasing clean energy credits for carbon offsetting [20]. Optimizing the electricity consumption of software within LLMs also holds potential, for instance, by enhancing model performance to reduce training time. The rise of hyper-scale data centers specifically for LLMs necessitates the development of more low-carbon technologies to cater to the immense computational demands. Meanwhile, existing evaluation and regulatory frameworks for AI should be adjusted to comprehensively encompass the unique characteristics of LLMs in areas such as large-scale computing and diverse applications.

LLMs, being a relatively nascent field, have prompted limited research on their available and sustainable electricity supply thus far. Table 1 outlines the current scope of literature in LLMs or related domains, highlighting a lack of comprehensive coverage across all pertinent topics. As such, to our knowledge, this review stands as the first to exclusively focus on estimating energy consumption throughout the lifecycle of LLMs, critically examining the associated estimation challenges, and offering comprehensive suggested solutions for fostering available and sustainable electricity for LLMs.

The major contributions are included the follows.

1. The electricity consumption from two essential stages of LLMs, i.e., training and inference, are systematically analyzed. Approaches and estimated results for leading models are summarized, providing a comprehensive perspective for understanding the electricity consumption characteristics of LLMs.

**Table 1**
Related literature and their coverages.

| Literature | Object of study | Stages of the LLM | Estimation approach | Challenges | Solutions |
|---|---|---|---|---|---|
| [10] | Broader AI | Training and inference | / | ☑ | ☑ |
| [11] | Many LLMs | Training and inference | / | ☒ | ☒ |
| [13] | Broader AI | Training and inference | / | ☑ | ☒ |
| [14] | Broader AI | Training | Measured | ☒ | ☑ |
| [16] | Specific LLM (LLaMA) | Inference | Measured | ☒ | ☒ |
| [18] | Specific LLM (BLOOM) | Training and inference | Measured | ☒ | ☒ |
| [21] | Specific LLM (GPT) | Inference | Measured and calculated | ☒ | ☒ |
| [22] | Many LLMs | Training and inference | Calculated | ☒ | ☒ |
| [23] | Broader AI | Training and inference | Calculated | ☑ | ☒ |
| [24] | Broader AI | Inference | Measured and calculated | ☑ | ☒ |
| [25] | Broader AI | Inference | Measured | ☒ | ☒ |

Notes: The symbol "/" indicates that neither of the approaches is introduced, "☑" indicates that the topic is covered, and "☒" indicates that the topic is not covered.

2. Given the challenges for electricity availability and sustainability of LLMs, a bottom-up framework with four major tiers is developed to elaborate on that affect LLMs' electricity, i.e., (i) software and hardware, (ii) data center, (iii) power grid, and (iv) external society. Both potential positive and negative aspects within each level are analyzed.
3. To achieve available and low-carbon electricity consumption, solutions are explored to improve LLMs' electricity supply from four dimensions as well, including: (i) improvement of software and hardware, (ii) application of low-carbon energy techniques, (iii) optimization of physical architecture, and (iv) enhancement of external evaluation and supervision.

The structure of this paper is organized as follows. Section 2 focuses on the models for estimating electricity demand by LLMs and comparing the evaluations with data from various literature sources. Section 3 delves into the primary challenges influencing the electricity consumption of LLMs, which also constitute significant barriers to accurate evaluation results. Section 4 suggests a combination of solutions that can enhance the availability and sustainability of electricity for LLMs. Section 5 presents the conclusions and outlook.

## 2. Evaluations of electricity demand for LLMs

Significant variations in electricity consumption and the corresponding carbon emissions exist among various LLMs, stemming from both distinct stages of training and inference, as well as inherent factor. In this section, the electricity demand within training and inference stages are evaluated separately by exploring influential factors and then by comparing diverse evaluation results.

### 2.1. Life-cycle stages of one LLM

To illuminate the power characteristics and further to mitigate their electricity demand, it is imperative to delve into the lifecycle stages of LLMs. A typical categorization focuses on their essential stages, i.e., training and inference.

The training stage develops a model by learning an underlying function through analyzing a dataset to select a set of parameters. The training commences with model development, where researchers design and assess model architectures [24]. Once a promising model is identified, it progresses to formal training, which may involve multiple iterations over the dataset for hyperparameter fine-tuning. Depending on the task requirements of LLMs, the model may consume intensive computational resources, requiring training for tens of thousands of hours on thousands of accelerators, which can be measured in terms of accelerator hours or accelerator years [26].

The inference stage involves deploying the trained model into a production environment to predict outputs for new queries. This is where the model is operationalized in the real world. For instance, given some prompts, trained LLMs classify these inputs and generate tailored outputs. A successful LLM may attract myriad users, producing billions of daily uses globally.

The distinct nature of the training and inference stages necessitates a separate consideration of their electricity consumption. A schematic overview of the differences of each stage is provided in Fig. 1.

Training constitutes an iterative process that involves batch processing of data. Consequently, it can leverage padding techniques to optimize the utilization of accelerators like GPUs, tensor processing units (TPUs), field programmable gate arrays (FPGAs), and neural processing units (NPUs) [27], thereby maintaining a relatively stable workload. Limited by bandwidth speed and latency, decentralized training remains highly challenging at present. As a result, training LLMs is more feasible in centralized data centers. Additionally, training workloads incorporate model checkpoints, enabling the resumption of training from interruptions in case of infrastructure failures or power outages. Moreover, fine-tuning, e.g., parameter-efficient fine-tuning, supervised fine-tuning for task-specific LLMs, can effectively reduce computational costs, and thus reduce energy demand [28].

In contrast, inference operates in real-time, as exemplified in applications such as chatbots that must promptly respond to dynamic user queries. Inference requests, unlike those of training, are unpredictable and cannot be planned in advance. Thus, accelerators for inference encounter idle periods between user requests, resulting in pronounced

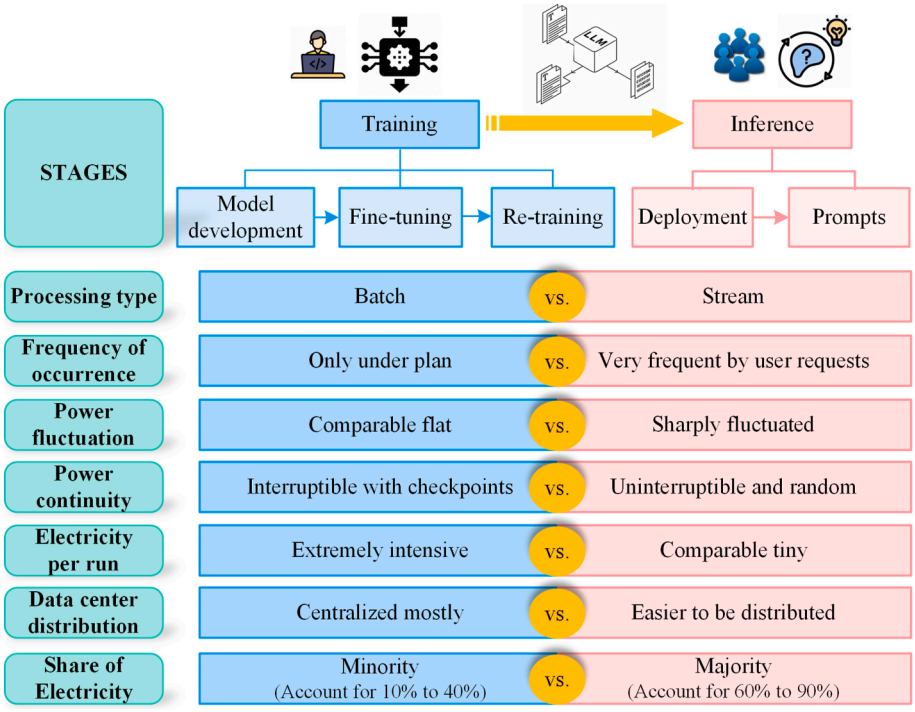



**Fig. 1.** Different characteristics between training and inference in one LLM's lifecycle.

power fluctuations. It is more feasible to deploy the inference process across different data centers and distributed devices. Moreover, inference workloads, often being business-critical, necessitate resilience strategies including uninterrupted power supply (UPS) systems (i.e., batteries) and geographical data center redundancies to ensure power continuity.

Training, while significantly more energy-intensive, occurs far less frequently than inference. Some studies have segregated two stages when evaluating the accumulated electricity consumption of LLMs. In Ref. [29], it is estimated that the current workload of AI, including LLMs, stands at 4.3 GW and may grow into 20 GW by 2028, with the share of inference increasing from 80 % to 85 %. In Ref. [30], Nvidia's workload is estimated to comprise 80 %–90 % of inference. Similarly, Amazon Web Services (AWS) assesses that inference also accounts for 80 %–90 % of total cloud computing demand [31]. Another prominent cloud computing provider and user, Meta, estimates that inference in LLMs is associated with 65 % of their carbon footprint [6]. One study from Google attributed 60 % of energy consumption to inference, while training accounted for 40 % [32]. Despite notable variations in these estimations, a consensus emerges that the total electricity consumed by inference will surpass that of training. This is because, although inference incurs very small power cost per execution, it occurs most frequently in the lifecycle of LLMs, thereby accumulates substantially and resulting in higher electricity demand than training [10].

### 2.2. Electricity demand evaluation for LLM training

#### 2.2.1. Factors for evaluating electricity consumed by training

The training process of LLMs is scheduled and energy-intensive, and thus attracts many attentions in the realm of evaluating LLMs' electricity demand. Despite the absence of standardized methodologies, the following factors have garnered widespread accordance.

The primary factor is the underlying model architecture. Presently, industry-standard LLMs are predominantly based on the transformer architecture [33], as indicated by the full names of GPT (i.e., Generative Pre-trained *Transformer*) and BERT (i.e., Bidirectional Encoder Representations from *Transformers*). This architecture primarily falls into two general categories, i.e., encoder-decoder and decoder-only. Given the transformer-based architecture and the vast amount of data involved, the training of LLMs is highly stable. Thus, the power fluctuation of the training process is comparable flat.

The second factor refers to the number of parameters. The neural scaling law has been fully validated in the current evolution of LLMs [34]. One of the core points is that there is a clear positive correlation between the performance of LLMs and the size of their training data, and the size of the training data directly and critically affects the number of model parameters. In natural language processing, tokens represent words or sequences of multiple characters, signifying the volume of data. The CORE dataset encompasses over 200 billion tokens, enabling the construction of an LLM with 10 billion parameters based on a recommended 20:1 token-to-parameter ratio [35].

Another hallmark of the scaling law is the direct proportionality between the operational complexity of training and the number of parameters [10]. The operational complexity is often measured in floating point operations (FLOPs), with the total FLOPs generally scaling quadratically with the number of parameters [36]. For instance, training a GPT model with 175 billion parameters requires about $3.7 \times 10^{24}$ FLOPs, which escalates to $1.2 \times 10^{26}$ FLOPs for a comparable model with 1 trillion parameters [37].

This escalating computational workload translates into a dual increase in both instant electric power and accumulated electric energy. Higher FLOPs correlate positively with greater electricity consumption, although the exact figures depend on software model architectures and hardware configurations, e.g., accelerators and central processing units (CPUs). Additionally, along with the increasing need of hyperparameter tuning for billion-parameter models, training hours extend significantly.

Finally, the power sources of the data centers used for training, combined with the total electricity, enables the estimation of carbon emissions. The carbon intensity of the power grid varies based on the power sources, with coal-fired grids emitting more carbon per kWh compared to hydroelectric or renewable-energy-powered grids [11]. Whether considering the PUE of data centers also affect the final result of the carbon emission analysis.

For convivence, some of these commonly used indicators above for evaluating electricity demand of LLMs are summarized in Table 2.

#### 2.2.2. Models for evaluating electricity consumed by training

Currently, two predominant approaches exist for evaluating consumed electricity during the training stage of LLMs, i.e., online measurement and offline formula-based estimation, as shown in Fig. 2.

The online measurement approach revolves around estimation using actual computing performance obtained directly from measurement, which is synchronized with the training process. For instance, the power consumption of GPUs can be ascertained by querying the NVIDIA's System Management Interface to sample power usages, while the power consumption of CPUs can be monitored through Intel's Running Average Power Limit interface [14].

Alternatively, the offline estimation approach is formula-based and thus can be complete theoretical, which enables predictions either prior to or subsequent to training [18]. The relationship between the size of their training data, number of parameters, operational complexity, training performance, and runtime, which can be approximated by the formulas in Fig. 2. The estimation formulas are mainly exemplified by the decoder-only transformer architecture, and adjustments to these calculations would be necessary for other structures. If multiple rounds of training are applied, the above formulas can be iteratively applied to obtain cumulative results.

#### 2.2.3. Estimations and comparisons for electricity consumed by training

*2.2.3.1. Estimated electricity consumption by training of various LLMs.* Leading LLMs encompass Google's BERT and PaLM, OpenAI's GPT

**Table 2**
Indicators for evaluating electricity demand of LLMs.

| Indicators | Abbreviation | Implication |
|---|---|---|
| Token | / | A piece of text, consisting of words utilized for natural language processing. For instance, in GPT models, 1 token is approximately equivalent to 4 characters or 0.75 words in English text. |
| Parameter | / | Weights in a neural network architecture, referring to the adjustable numerical values that define the capabilities of the model. |
| Floating point operations | FLOPs | Operational complexity of a certain task, mainly depending on software (e.g., LLM architecture, number of parameters). |
| Floating point operations per second | FLOPS or FLOPs/second | Computing performance by a certain equipment, mainly depending on hardware (e.g., processor types). |
| Power Usage Effectiveness | PUE | The industry-recognized benchmark for assessing efficiency of data centers, by calculating the ratio of total electricity consumption (including indirect costs such as cooling and lighting) to the direct electricity utilized by computational hardware. |
| $CO_2$ equivalent emissions | $CO_2$eq | $CO_2$ and all the other greenhouse gases (e.g., methane, nitrous oxide), measured in units such as ton or kg. |
| Carbon intensity | / | The cleanliness of the electricity of a power grid or certain power supply infrastructure, measured in units such as $kgCO_2eq/kWh$ or $tCO_2eq/MWh$. |

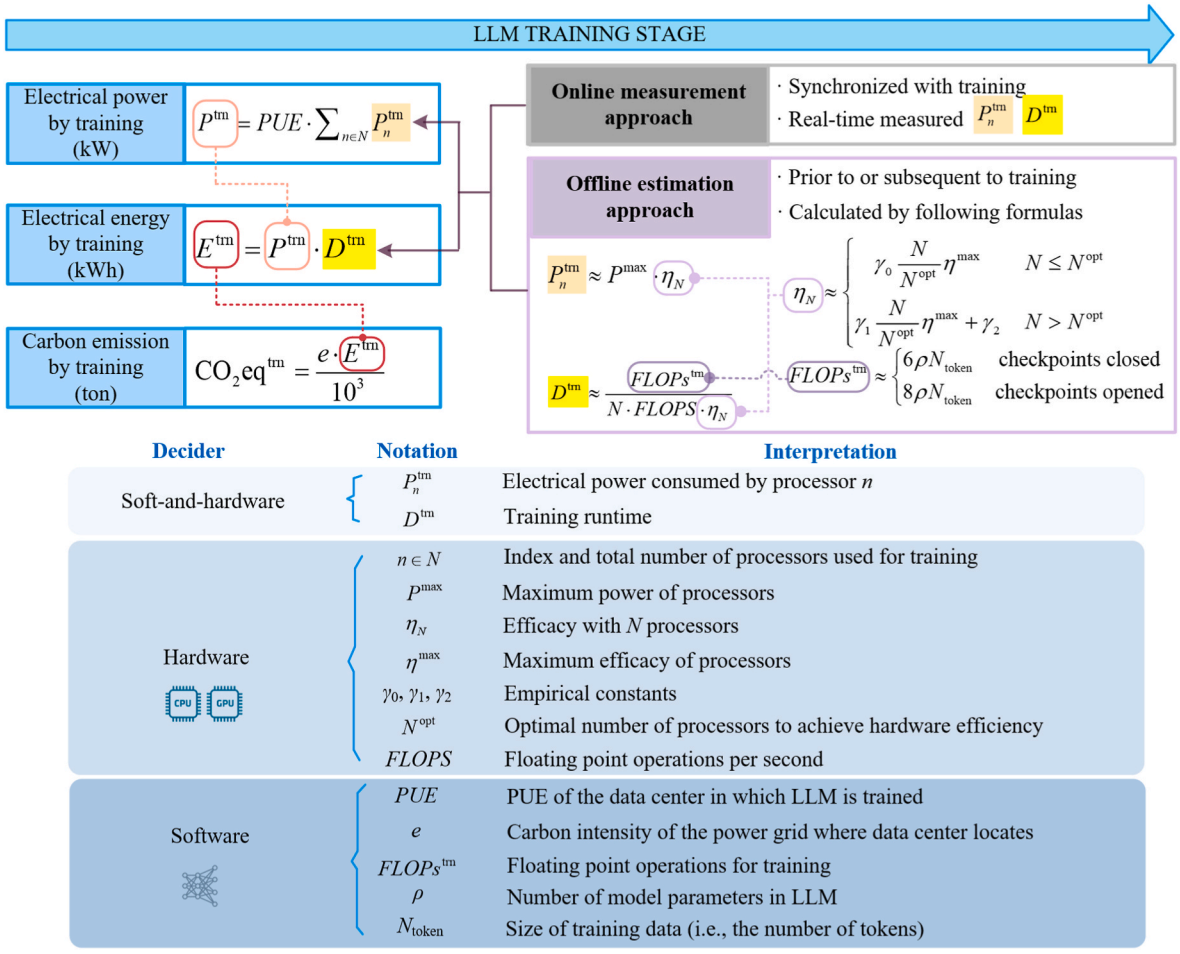



**Fig. 2.** Key formulas and parameters for evaluating electricity demand of training.

**Table 3**
Parameters and electricity consumption evaluations for several leading LLMs.

| Developer | Model | Released year | Number of parameters (billion) | Accelerator type | Electricity consumption (MWh) | $CO_2$ emissions (ton) | Literature |
|---|---|---|---|---|---|---|---|
| Google | BERT | 2018 | 0.1 | GPU V100 | 1.5 | 0.7 | [13,14] |
| | GLaM | 2021 | 1162 | TPU v4 | 456 | 40 | [19] |
| | PaLM | 2022 | 540 | TPU v4 | 3436 | 271.4 | [38] |
| OpenAI | GPT-2 | 2019 | 1.5 | TPU v3 | 1.7 | 0.7 | [14,39] |
| | GPT-3 | 2020 | 175 | GPU V100 | 1287 | 552 | [32,40] |
| | GPT-4 | 2023 | 1800 | GPU A100 | 51,772 to 62,318 | 12,456 to 14,994 | [41,42] |
| Meta | OPT | 2022 | 175 | GPU A100 | 324 | 75 | [43] |
| | LLaMA 2 | 2023 | 7 | GPU A100 | 74 | 31.2 | [44] |
| | | | 13 | | 147 | 62.4 | [44] |
| | | | 70 | | 688 | 291.4 | [44] |
| | LLaMA 3 | 2024 | 1 | GPU H100 | / | 107 | [45,46] |
| | | | 3 | | / | 133 | [45,46] |
| | | | 8 | | / | 390 to 420 | [45,46] |
| | | | 70 | | / | 1900 to 2040 | [45,46] |
| | | | 405 | | / | 8930 | [45,46] |
| BigScience | BLOOM | 2022 | 1 | GPU A100 | 159 | 9.1 | [18] |
| | | | 104 | | 267 | 15.2 | [18] |
| | | | 176 | | 433 | 24.7 | [18] |
| DeepMind | Gopher | 2021 | 280 | TPU v3 | 1151 | 380 | [13,47] |
| | | | | / | 1066 | 352 | [4] |
| TII | Falcon | 2023 | 40 | GPU A100 | 163[b] | 8 to 90[b] | [48–50] |
| | | | 180 | | 2,940[b] | 147 to 1,617[b] | [48–50] |
| Mistral AI | Mixtral 8 × 7B | 2024 | 47[a] | GPU A100 | 232[b] | 6.3[b] | [51–53] |
| DeepSeek | DeepSeek-V3 | 2024 | 671[a] | GPU H800 | 1,025[b] | 550[b] | [54,55] |

[a] These values denote the total number of parameters, with different quantities activated depending on the model architecture.
[b] The electricity consumption data of some LLMs lack direct values from open-published literature. Thus, these data are estimated, and the estimation process is provided in Appendix Table A1.

series, Meta's OPT and LLaMA series, Anthropic's Claude series from the U.S., BigScience's BLOOM and Mistral AI's Mistral series from France, TII's Falcon series from the UAE, DeepSeek's DeepSeek series from China, *etc*. Their key parameters and representative electricity consumption evaluations are outlined in Table 3, and comparisons reveal several common traits as follows.

1. Typically, larger models consume more electricity in the same series. This trend is notably evident in both Meta's LLaMA2 and LLaMA3 series, in which all trained on the same data center with the same type of GPU. For LLaMA2 series, as the number of parameters increases from 7 billion to 70 billion, the consumed electricity also increases by nearly 10 times. Data from LLaMA 3 not only show a positive correlation but also reveal the non-linear growth between parameter scale and electricity consumption. This pattern suggests that efficiency optimizations may partially mitigate the energy impact of scaling, though not sufficiently to offset the overall growth.
2. While larger models generally require more power than smaller ones, the exact figures vary significantly between models. For instance, GPT-3 with 175 billion parameters generated about 552 tons of $CO_2$eq, whereas similarly sized models like OPT with 175 billion parameters and Gopher with 280 billion parameters exhibit notably smaller carbon emissions.
3. Computational efficiency of training LLMs is continually improving over time. For instance, 18 months after GPT-3, Google introduced GLaM, an LLM with 1.162 trillion parameters. Despite being nearly seven times larger and outperforming other LLMs, GLaM's training consumed 2.8 times less electricity. This improvement results from both advancements in training techniques and energy efficiency optimizations of accelerators. Compared to contemporary accelerators, the energy-efficient TPU v4 used to train GLaM consumes only about one-sixth to one-half of power [56].

*2.2.3.2. Carbon intensity and their impact for LLM training.* Besides the overall consumed energy, the carbon footprint is also directly shaped by the carbon intensity of power supply infrastructure where the LLM is trained. Fig. 3 offers an intuitive comparison of three LLMs from different countries in Table 3 to show the impact of carbon intensity. GPT-3 trained on the U.S. grid and DeepSeek-V3 trained in China, show a roughly 1.26-fold difference in electricity consumption, yet their carbon footprints differ by only about 0.4 %. This near-parity in emissions arises because China's average grid carbon intensity, though higher than that of the U.S., results in a negligible gap when paired with DeepSeek-V3's lower energy use.

In contrast, BLOOM with 176 billion parameters, consuming about one-third of GPT-3's electricity, achieves a remarkable 22-fold reduction in its carbon footprint. This significant difference is attributed to France's abundant nuclear electricity supply, which can obtain a carbon intensity at least 7.5 times lower than GPT-3 and 10.5 times lower than DeepSeek-V3. The carbon intensity used for BLOOM in Fig. 3 is adopted from Ref. [11], and if the value from Ref. [53] were used, this emission gap would widen further.

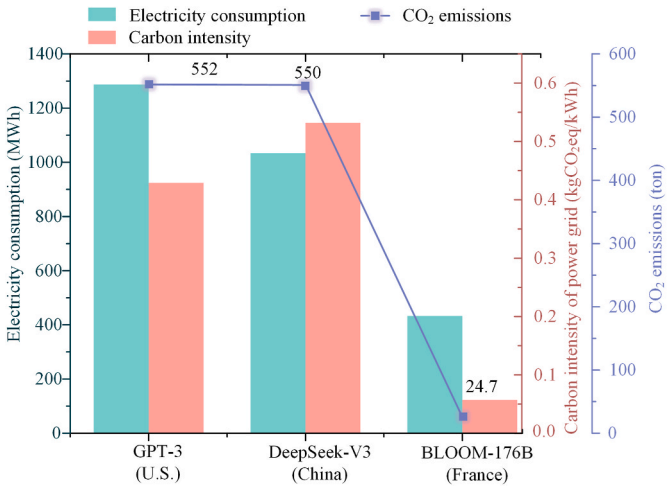



**Fig. 3.** Impact of carbon intensity on emissions for three representative LLMs.

These real-world cases confirm that carbon intensity is a critical determinant of an LLM's energy sustainability, often outweighing raw energy consumption in shaping environmental impact.

*2.2.4. Power requirements of fine-tuning*

Fine-tuning has emerged as a critical sub-stage in the training process. This involves optimizing pre-trained LLMs from domain-specific datasets to efficient hyper-parameters, differing from full training in both computational scope and energy demands. This subsection examines the energy demand of fine-tuning, comparing with those of full training and inference.

Compared to full training or training with pre-training, fine-tuning typically consumes far less energy. This disparity stems from several key factors, e.g., reduced dataset size, selective parameter optimization, and shorter training duration. Fine-tuning bypasses initial training stages, instead, it efficiently adapts pre-trained LLMs to downstream applications, significantly lowering computational resource demands and costs [57]. A case study on a LLM named Munin with 7 billion parameters showed that one epoch of fine-tuning took 4 h, consumed 0.694 kWh of electricity, and emitted 57 g $CO_2$eq [58], which can be even neglected compared to the energy required by full training.

When compared to inference, fine-tuning consumes more energy per task, but its cumulative energy impact differs across applications. It is founded in Ref. [59] that the power intensity of fine-tuning varies significantly with sequence length, of which the inference is relatively stable. Additionally, fine-tuning generally requires higher energy than inference because it demands more computational resources and time to update hyper-parameters [60].

Table 4 compares the energy demands by fine-tuning relative to full training and inference. Overall, fine-tuning serves a critical tool to balance energy efficiency and model adaptability. While its energy demands are minimal compared full training or pre-training, frequent or large-scale fine-tuning can accumulate significant energy consumption, particularly in sectors requiring continuous model updates.

### 2.3. Electricity demand evaluation for LLM inference

*2.3.1. Factors affecting electricity consumed by inference*

LLMs' electricity demand does not end with training. After training, inference serves as a tangible end-to-end application of LLMs, with pivotal factors influencing its electricity demand, including LLM itself along with deployed model size, modality tasks, and end prompts [1, 16].

Firstly, the model size is a prominent contributor for consumed. As shown in Fig. 4, the average electricity per inference varies significantly across model sizes, exemplified by the comparison of BLOOM and Flan-T5 under different numbers of parameters. Regardless of the specific

**Table 4**
Comparisons of energy demand by fine-tuning vs. training and inference.

| Stage | Power intensity | Time duration | Energy consumption |
|---|---|---|---|
| Pre-training | Intensive | Days to weeks | High |
| Training | Intensive | Months | Extremely high |
| Fine-tuning | Medium | Hours to days | Moderate (far less than training) |
| Inference | Low | Milliseconds to seconds | Depending on tasks |

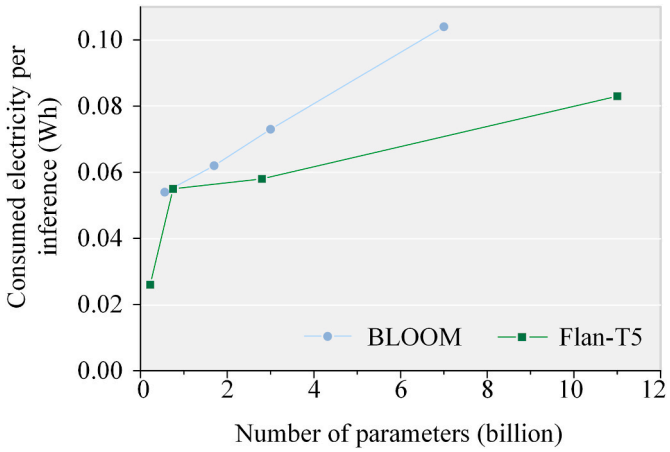



**Fig. 4.** Distributions of consumed electricity per inference under different numbers of parameters by two LLMs.

LLM, a clear positive correlation emerges between model size and electricity used for inference [11].

Secondly, different modality tasks exert an even more pronounced influence on electricity demand. Common modalities in LLMs include single specific tasks (e.g., text classification, text-to-text generation) as well as multimodal tasks (e.g., image captioning). These modalities lead to varying requirements for hardware and differing computation durations, which in turn affect both the instant power consumption and the accumulated energy usage.

Additionally, the length and frequency of prompts by users significantly contribute to the power usage [16]. The number of prompts processed within an evaluation duration directly correlates with energy consumption. During the 18-day inference demonstration in Ref. [18], the power used for inferences fluctuated between 1.252 kW and 2.735 kW. This notable variation, which is significantly greater than that observed during training, is attributed to the unpredictable inference requests. Moreover, it underscores the potential for substantial errors when estimating electricity demand for inference based solely on the maximum power drawn during training, since GPUs remain idle in between users' prompts.

Finally, the estimation of carbon emissions also relies on the power supply mix of data centers, which is similar to the training stage. But differently, the cumulative energy consumed by inference can be distributed across geographical regions and years.

### 2.3.2. *Models for evaluating electricity consumed by inference*

The electricity demand associated with inference, in contrast to that of training, has garnered relatively little attention [11], and suffer from a lack of standardized, formulaic approaches [4]. This is partly due to the unique characteristics of inference, including its tiny per-instance energy consumption, diverse task types, and the volatility stemming from fluctuating usages, as shown earlier in Fig. 1. Another contributing factor lies in the significant variation in electricity across different inference tasks, deployed on varying LLMs and processors. A summary of current models for estimating the electricity and carbon emission for inference is presented in Fig. 5.

Given the negligible electricity per task, current evaluations for inferences tend to emphasize electrical energy over power. Under a certain task type of one LLM, the electrical demand of a task typically correlates positively with its operational complexity, the response time per token, and the token length of the prompt. The decentralized nature of inference activities allows for assessments across different locations. Data

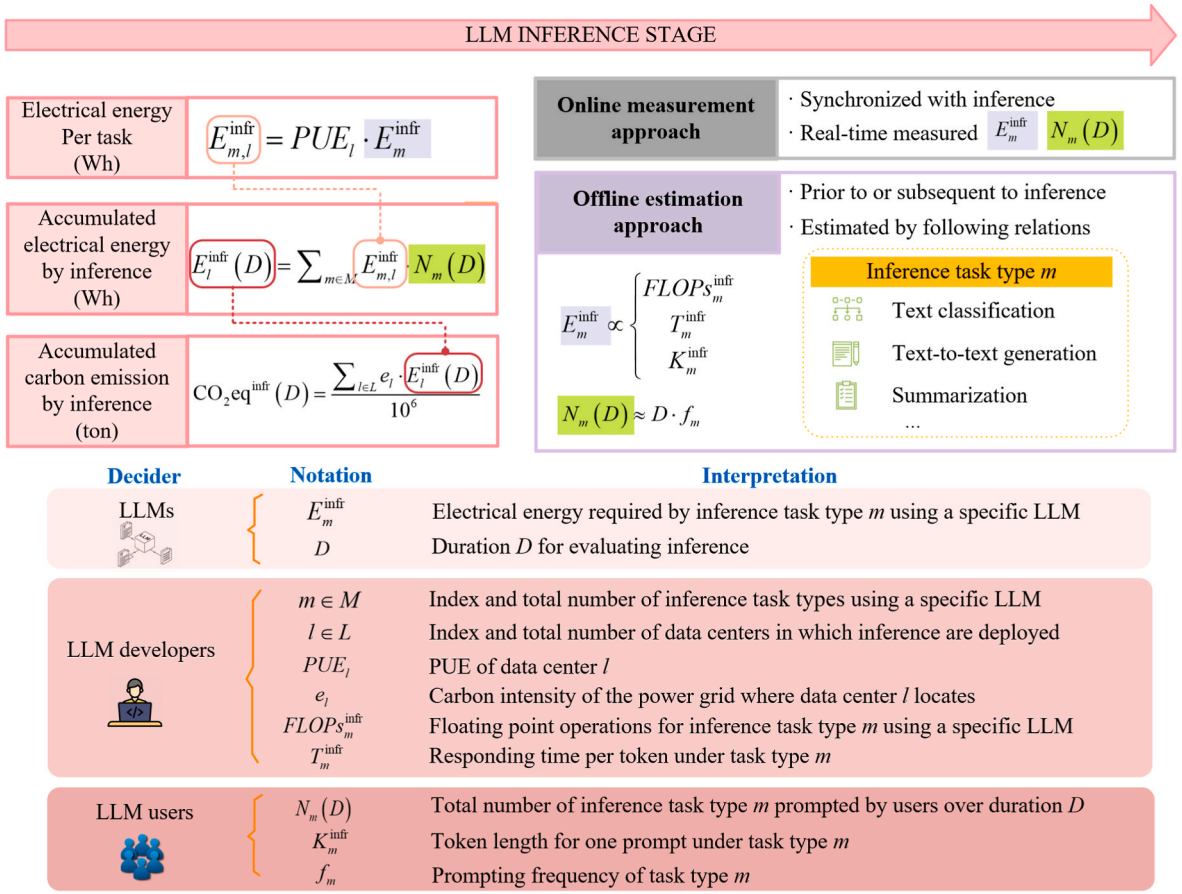



**Fig. 5.** Key formulas and parameters for evaluating electricity demand of inference.

centers located in various regions exhibit varying PUEs and carbon intensities, which can be individually incorporated when estimating energy consumption and carbon emissions.

However, unlike training, inference lacks a systematic, formula-based estimation framework [16,21]. Data for these parameters are often generated from statistics. And differing from the one-time nature of training, the electricity consumed by inference is determined by the evaluation duration. When numerous users generate massive prompts, the cumulative energy expenditure from these minute power instances becomes substantial.

#### 2.3.3. Estimations and comparisons for electricity consumed by inference

The prevalent classification adopted herein segregates electricity consumption based on task types. Table 5 presents the average electricity and its standard deviation for various models across distinct inference tasks, i.e., Einfr $m,l$ in Fig. 5 [4,11]. Several common traits emerge as follows.

1. Significant variances lie in electricity requirements across tasks. In general, single specific tasks around classification exhibit lower power levels, while single specific tasks like text generation fall in the middle range, and multi-modal tasks occupy the highest tier. Specifically, text-to-image generation consumes over 60 times the energy used by text-to-text generation (2.907 Wh vs. 0.047 Wh). The disparity in energy spans from a minimal 0.002 Wh per inference for text classification to an average peak of 2.907 Wh for text-to-image generation, representing a staggering over 1450-fold difference.
2. Although less pronounced than task type, an increase in the number of parameters in LLMs typically correlates with higher electricity demands during inference. For instance, the most carbon-intensive image generation model, i.e., stable-diffusion-xl-base-1.0 with 3.5 billion parameters, emits 1.594g of $CO_2$eq per inference, whereas a smaller model, i.e., segmind/tiny-sd with 500 million parameters, emits only 0.1g [11]. This variance contributes to the significant standard deviations observed in Table 3.
3. The consumed electricity during the inference stage will be far exceed that during the training stage under accumulation. This relationship is referred to as cost parity in, where the number of inferences required to equal the training cost varies inversely with the individual energy consumption per task. Currently, ChatGPT boasts over 10 million daily active users with an estimated daily energy exceeding 564 MWh [61,62]. The training stage of GPT-3 is estimated to have consumed 1287 MWh. Consequently, it can be inferred that within merely 2.28 days, the electricity consumed by inference would surpass that of training.

### 2.4. Discrepancies in estimations and underlying reasons

Existing estimations on LLM energy consumption, including both training and inference stages, exhibits substantial discrepancies, even for the same model. Examples include the BERT case in the introduction, the values for Gopher in Table 3, and the widely cited inference energy data [4,11] that contrasts sharply with a new study [25], which reports that recent released models like GPT-o3 and DeepSeek-R1 consume far more energy than previously estimated. These inconsistencies stem from three key reasons.

**Table 5**
Means and standard deviations of consumed electricity per prompt under different inference tasks.

| Modality | Task type | Mean (Wh) | Std (Wh) |
|---|---|---|---|
| Single | Text classification | 0.002 | 0.001 |
| | Extractive question answering | 0.003 | 0.001 |
| | Masked language modeling | 0.003 | 0.001 |
| | Token classification | 0.004 | 0.002 |
| | Image classification | 0.007 | 0.001 |
| | Object detection | 0.038 | 0.02 |
| | Text-to-text generation | 0.047 | 0.03 |
| | Summarization | 0.049 | 0.01 |
| Multi-modal | Image captioning | 0.063 | 0.02 |
| | Text-to-image generation | 2.907 | 3.31 |

Firstly, studies vary significantly in how they define the boundaries of energy measurement. Some account for data center overheads (e.g., cooling, power distribution), while others focus exclusively on specific computing hardware. In Ref. [18], 100 % GPU occupation is assumed for BLOOM's training, while the CPU consumption is neglected. Estimates may also isolate training stages or exclude intermediate training rounds. Furthermore, many LLMs rely on cloud servers for training, (e.g., Mistral AI models using AWS) and open-source local deployment for inference, complicating direct estimates.

Secondly, hardware accelerators and model configurations collectively drive statical variations. Significant efficiency disparities exist across accelerator generations. For instance, training BERT on a 2018 V100 GPU consumes 2.1 times more energy than on a 2022 H100 GPU, while hardware differences between A100 and H200 account for nearly 20 % of energy variance in GPT-4o mini's inference [63]. Model-specific settings, including pre-training scale, task-specific fine-tuning intensity, and activated parameters, directly modulate computational loads, consequently amplifying estimation discrepancies. For instance, BLOOM's intermediate 1-billion, 13-billion, and 104-billion models contributed 35.8 tons of $CO_2$eq, exceeding the final model's single training emission.

Thirdly, geographic and operational contexts introduce variability. Even if standardized electricity measurements emerge, detailed rules will be needed to ensure uniform carbon footprint calculations as well. Carbon intensity depends not only on the regional grid's location but also on a data center's energy management strategies concerning and local renewable and UPS utilization. Furthermore, current carbon intensity values published by providers like AWS incorporate carbon offsets, which represent compensatory measures rather than actual real-time consumption. No standardized framework currently governs this practice, thereby compromising the comparability of estimations.

In summary, significant discrepancies in LLM energy estimations stem collectively from methodological inconsistencies in measurement boundaries, configuration-driven computational variations, and operational carbon intensity factors. These factors collectively highlight the need for evaluation standardized frameworks toward reliable estimates across different models.

## 3. Challenges influencing electricity demand of LLMs

Accurately evaluating electricity demand by LLMs is fraught with challenges. This section delves into the multifaceted obstacles and overcoming, which will shape the actual electricity consumption and carbon footprint of future LLMs. Fig. 6 encapsulates these challenges, categorizing them into four major tiers ascending from foundational to more intricate levels: (i) software and hardware, (ii) data center, (iii) power grid, and (iv) external society. Some of these challenges act as constraints, limiting the expansion of LLMs and their power demand, while others stimulate their rapid progress. Consequently, these diverse influences contribute to the inherent uncertainties surrounding LLMs' electricity usage and the associated carbon emissions.

### 3.1. Improving software and hardware architectures

As the number of parameters and computational demands of LLMs boom, advancements in the efficiencies of software and hardware systems are accelerating in parallel [64]. With LLM products iterations, these systems become increasingly tailored to LLM-specific applications, further optimizing their efficiencies [24]. Consequently, the long-term trend in LLMs' electricity demand remains uncertain.

Some perspectives argue that, driven by the neural scaling law, the

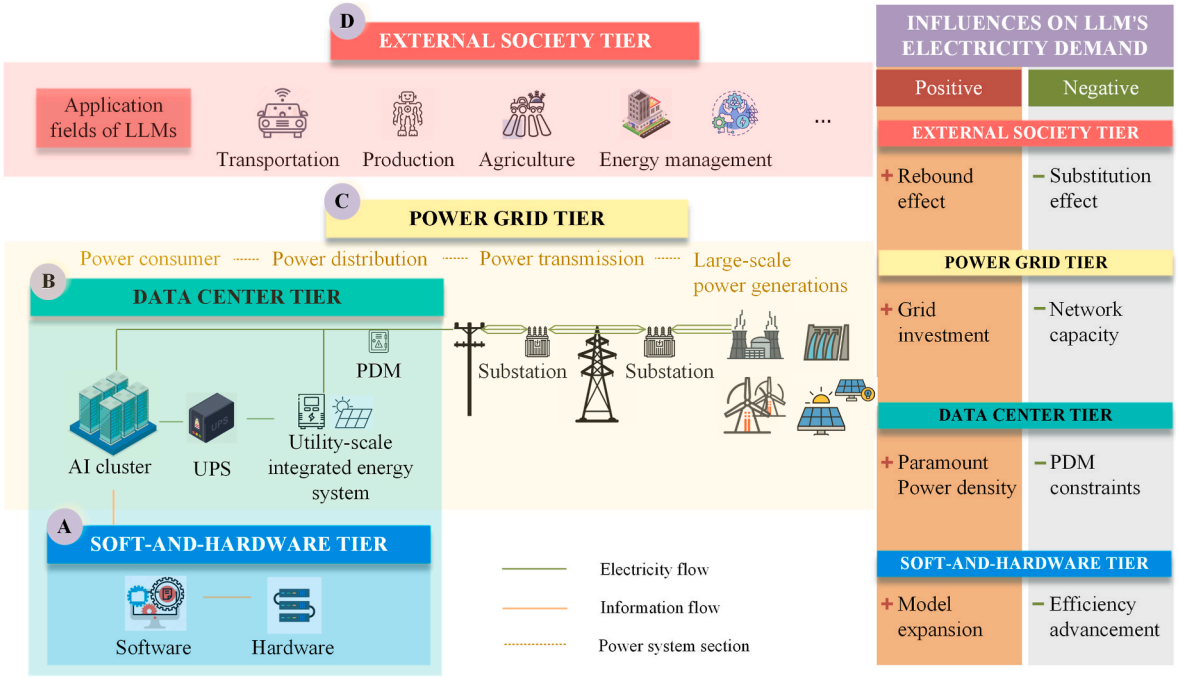



**Fig. 6.** Bottom-up framework with four tiers of challenges and their factors influencing electricity demand of LLMs.

electricity usage of LLMs will inevitably surge in tandem with model expansion. Both the individual size and total quantity of LLMs are expanding, and their iterations are accelerating. Reference [65] suggests that halving errors in LLMs necessitates a staggering 1.9-million-fold increase in computational resources.

Conversely, other viewpoints contend that advancements in hardware and software, e.g., model architectures and algorithms, could significantly enhance efficiency, potentially stabilizing or even reducing LLMs' energy demands [66]. These studies arrived at this conclusion by drawing analogies with the growth trends in traditional data center demands. For instance, between 2010 and 2018, global data centers witnessed a 550 % increase in computing power, yet their electricity consumption only rose by 6 % [67]. Similar trends have already emerged in the field of LLM.

Beyond advancements in locally deployed software and hardware, the deployment paradigms of LLMs are also undergoing transformations. A foreseeable trend is the increasing collaboration between edge devices and cloud servers. For instance, on-device learning at the edge mitigates latency, reduces reliance on network connectivity, and enhances data privacy. Reference [68] introduced an LLM inference engine tailored for mobile phones. However, edge devices often face challenges in accessing renewable energy sources and accurately estimating their carbon footprint [23]. On the other hand, they may offer higher device utilization rates [69]. As a result, the impact of cloud-edge deployment on the overall electricity demand and carbon emissions of LLMs remains indeterminate [19].

### 3.2. Extreme power density in data centers

To cater to the exponential surge in computational demands of LLMs, the necessity of sophisticated infrastructure support is paramount. The rapid evolution and application of LLMs are reshaping the electrical design of modern data centers. Distinguishing from conventional data center requirements, LLMs necessitate hyper-scale data centers capable of handling immense volumes of data storages and computations. This computational explosion, in turn, intensifies power consumption.

LLM training workloads consistently operate at exceptionally high densities. These workloads are organized into so-called AI clusters, consisting of racks of data storage units and computing processors, whose numbers varies based on sizes of data and parameters. Meanwhile, to handle inference workloads, as the number of hardware racks escalates, so does the demand for hardware racks. Popular LLM applications may necessitate several times more racks for inference than for training.

The massive cluster scale, consistently high workload utilization, and the substantial thermal design power (TDP) requirements of accelerators pose pressure on the physical infrastructure. The traditional size of power distribution modules (PDMs) is now inadequate for current capacities of these LLM-serving data centers. Only one decade ago, a 300 kW PDM could support 100 racks (averaging 3 kW per rack). Today, NVIDIA's H100 chip, connecting 256 devices at 700 W each, necessitates at least 179.2 kW per rack [70]. The Memphis Supercluster, constructed by xAI in July 2024, has integrated 100,000 H100 chips [71]. Moreover, in some regions, e.g., North America and Japan, where lower voltage standards (120 V or 100 V compared to 220–240 V in other regions) prevail, escalate current levels, limiting the number of circuits within high-density racks.

In other words, as the scale of hardware expands, the power density of data centers becomes increasingly extreme. On the other hand, the physical constraints from infrastructure hinder further increases in data center power density, ultimately constraining the upper limit of the electrical capacity these facilities can sustain.

### 3.3. Stressed and carbon-intensive energy supply from power grid

The extreme power density of data centers not only challenges their own PDMs, but also imposes significant pressure on their outer power distribution networks. Unlike the AI industry, the energy sector necessitates long-term capital investments with payoffs that do not materialize swiftly. And the delivery of additional power supply lags significantly behind the rapid expansion of LLM-serving data centers. Industry leaders from OpenAI, Meta, Tesla, and others concur that the constraint on LLM development is transitioning from silicon scarcity to electricity shortages [72,73].

The shortage of power supply in distribution networks is emerging as a critical energy bottleneck hindering the sustainable development of

LLMs. In regions with dense concentrations of data centers, such as Northern Virginia in the US, Beijing in China, London in the UK, and Frankfurt in Germany, local power grids already confront severe imbalances in power supply and demand, as well as challenges to grid stability. These issues limit the sustainable power support necessary for further expanded capacities of data centers. In response to these challenges, some regions have proposed levying special tariffs. For instance, the state of Ohio in the US has introduced a new tariff structure that mandates data centers to pay for at least 90 % of their contracted electricity usage, regardless of actual consumption [74].

Meanwhile, this escalating power demand of LLMs parallels their burgeoning electricity consumption. As energy consumption grows substantially, the economically-efficient and carbon-neutral objectives of running data center become increasingly difficult. For instance, a data center of Microsoft Azure can consume 11,870 kWh per day. The International Energy Agency predicts that the electricity demand of data centers in the EU will increase by 30 % in 2026 compared to 2023 [75]. By 2024, the electricity consumed by China's data centers are projected to reach 380 billion kWh annually [76], resulting in substantial electricity bills. When considering the adoption of green energy to meet zero carbon targets, such as purchasing renewable energy credits or carbon offsets, additional costs related to electricity are incurred. Given that LLMs have yet to establish mature business modes, these rising electricity-related expenses threaten to constrain their commercial expansion.

Furthermore, the environmental impact of LLMs, spanning training and inference stages, is closely linked to the carbon intensity of different local electricity grids and the energy construction of their power supply powering their computational infrastructure. Currently, over 15 % of global data center power capacity is concentrated in the top ten markets [77]. Fig. 7 shows their capacities and the carbon intensities of their respective local power grids [78–80]. It highlights a critical issue that major computational hubs with high capacities, such as Beijing with 2.3 GW and Shanghai with 2.0 GW, often correspond to regions with high carbon intensity, i.e., 0.645 and 0.690 $kgCO_2eq/kWh$. Consequently, LLM training and inference in such regions result in carbon emissions that are 3–5 times higher than those in areas with lower carbon intensity, such as London and Dublin. This disparity underscores the need for grid-aware LLM deployment strategies.

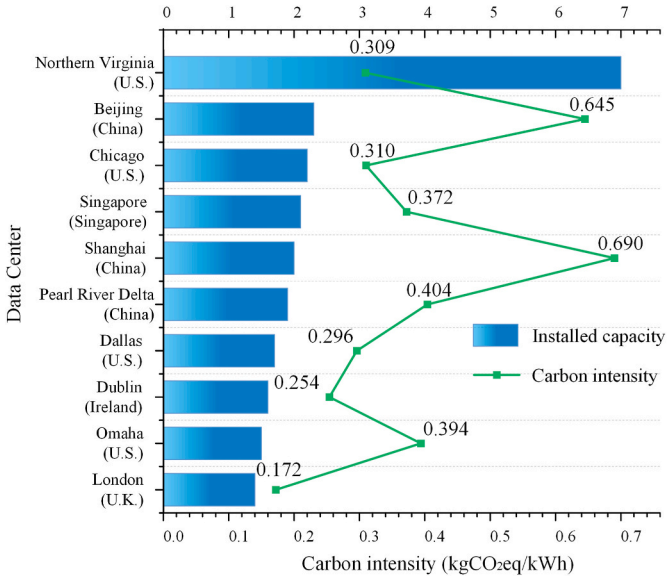



**Fig. 7.** Power capacity and carbon intensity of top ten data center markets.

### 3.4. Social externalities as rebound effect and substitution effect

LLMs, as a technological application, elicit numerous extraneous impacts across various industries. While these impacts do not directly alter their inherent energy efficiency, they influence the overall electricity consumption of LLMs through the rebound effect and the substitution effect [81].

When deployed to reduce costs in emission-intensive activities, LLMs may inadvertently stimulate increased consumption of these activities, triggering a structural *rebound effect* that elevates both electricity usage and carbon emissions by LLMs [82]. For instance, LLMs can enhance the performance of autonomous vehicles, which in turn increases vehicle mobility rates, further driving the adoption of LLMs [83]. For another instance, LLMs can aid in addressing challenges related to the safe and stable operation of power systems, and simultaneously, protectors of these systems may employ comparable LLMs for preventive measures [84]. Similar applications may also occur in various energy management systems [85]. These factors collectively have the potential to impact the overall utilization of LLMs.

On the contrary, despite perceptions of LLMs contributing to additional electricity and environmental costs, the *substitution effect* significantly enhances their sustainability [10]. LLMs outperform humans in tasks such as image and text generation, thereby reducing carbon emissions [86]. AI demonstrates immense potential in mitigating emissions across transportation, agriculture, and energy sectors [87]. In Ref. [88], LLMs like GPT-4 and Claude-2.1 facilitate the creation of engineering decision support systems through simple prompts, eliminating the need for complex programming. Models like Meta's LLaMA enable efficient analysis and processing of power reports [89]. These advancements lower the barriers to AI adoption, empowering enterprises and governments to devise greener policies and promote energy-saving technologies [90].

The widespread adoption of LLMs may perturb political and economic stability, eliciting ripple effects on poverty, food security, and social inequalities [91]. While developers strive to enhance LLM efficiency to minimize electricity consumption, such advancements often augment energy demand, resulting in a net increase in energy utilization by LLMs.

## 4. Solutions towards electricity availability and sustainability for LLMs

The electricity for LLMs can be discerned from two perspectives: the capacity of sufficient electricity supply (i.e., electricity availability) and the capacity for green electricity supply (i.e., electricity sustainability), where the former is the foundation of the latter. These challenges analyzed in Section 3 not only impact the electricity demand evaluations of LLMs but also pose barriers to their development. Pivotal solutions necessitate a holistic approach that originates from and transcends physical entities as well, encompassing software and hardware designs, physical architectures, low-carbon energy strategies, and external assessment and supervision, as illustrated in Fig. 8.

### 4.1. Improvement of software and hardware for LLMs

#### 4.1.1. Efficiencies of models and platforms

Improving the model architecture of LLMs is a key direction for mitigating electricity demand [92]. For instance, within the decoder-only framework, the incorporation of the Mixture-of-Experts (MoE) architecture effectively enhances model capacity while reducing computational demands, thereby decreasing power usage [13]. Accompanied by architectural advancements, a variety of optimization tools have also emerged, collectively enhancing efficiency and thereby reducing energy consumption.

The comparisons among GPT-3, GLaM, and Mixtral-8x7B shown in Table 6 highlight the efficiency gains under this trend [93]. Equipped

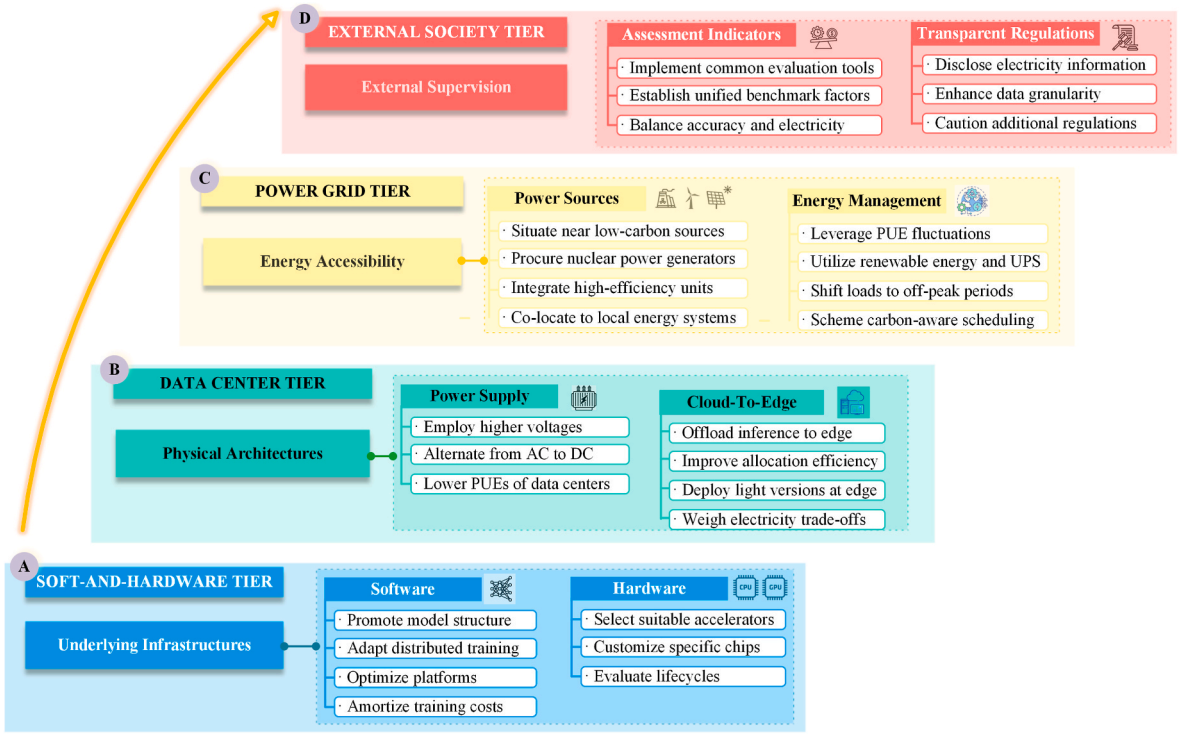



**Fig. 8.** Solutions towards available and sustainable electricity for LLMs.

**Table 6**
Comparisons among GPT-3, GLaM, and Mixtral 8 × 7B.

| LLM name | GPT-3 | GLaM | Mixtral 8 × 7B |
|---|---|---|---|
| Release year | 2020 | 2021 | 2023 |
| Number of parameters (billion) | 175 | 1162 | 47 with 13 activated |
| Model architecture type | dense | MoE | Sparse MoE |
| Accelerator type | GPU V100 | TPU v4 | GPU A100 |
| TDP (W) | 300 | 175 | 400 |
| Accelerator years | 405 | 144 | 63 |
| Electricity consumption (MWh) | 1287 | 456 | 232 |

with MoE, GLaM consumes merely 2.8 times the accelerator years of GPT-3, while boasting a sevenfold increase in parameter size compared to GPT-3 and achieving superior generative capabilities [19,32]. The electricity used to train GLaM is one-third that of GPT-3. Only two years later, Mixtral 8 × 7B further refined its efficiency, thereby achieving even greater energy reduction. By utilizing a Sparse MoE architecture and optimized through fine-tuning and direct preference optimization, Mixtral 8 × 7B matches the performance of GPT-3.5 while using far fewer parameters and consuming only 18 % of the training electricity required by GPT-3.

Advancements in distributed training of LLMs have also introduced multifaceted techniques to expedite training and minimize energy demand. Key technologies include pipeline parallelism [94], tensor parallelism [95], memory footprint reduction [96], low-precision computations [97], and *etc*. For instance, to conserve electricity in the LLM called Yuan 1.0, techniques in Ref. [98] involves incorporating enlarged hidden layers and optimized micro-batch sizes. The zero redundancy optimizer (ZeRO) proposed in Ref. [99] augments data parallelism by minimizing gradient, optimizer state, and offloaded memory usage. Falcon series, originated from the UAE, employs a 3D parallelism approach integrated with ZeRO, achieving world-leading performance with high efficacy [48].

Optimizing software platforms is another crucial pathway to enhancing power efficiency [100–102]. In a specific LLM task in Facebook, i.e., language translation, by leveraging pre-computation and caching mechanisms, the frequently accessed embedding vectors for translation tasks are efficiently managed, achieving a 6.7-fold improvement in energy efficiency [23]. By reducing iterations for hyper-parameter fine-tuning, direct energy savings are achieved [103,104]. Judicious reduction in network complexity simplifies models and halves overall electricity used by training [101,105,106].

Customized amortization of training costs is another strategy. Fine-tuning existing LLMs into develop task-specific ones can yield tailored versions, enhancing energy efficiency through increased reuse and partially amortizing training costs [107]. This pattern can be observed in OpenAI's progression from GPT-4 to GPT-4.0 and subsequently to GPT-4o mini. An open-source tool named xTuring in Ref. [108] enables developers to fine-tune popular LLMs with specified datasets for customization. By caching responses to common prompts instead of generating new responses for each instance, computational resources can be conserved. A learned linear growth operator strategy proposed in Ref. [109] can convert smaller models into larger ones, potentially reducing computational costs by about 50 %, thereby lowering power expenses.

Some LLM-focused software efficiency enhancements are validated in industry deployments, as shown in Table 7 [110–113]. By improving training efficiency and reducing training time, these techniques not only enhance performance of LLMs but also lower their overall energy demand.

Table 8 focuses several measures in Table 7 by taking DeepSeek series as a practical application [114,115]. This progression reveals rapid technical evolution. It only takes three months from DeepSeek-V1, which adopted baseline LLaMA 2, to DeepSeek-V2 with MoE and MLA architectures, which reduced training costs to 42.5 % of DeepSeek-V1. DeepSeek-V3 further revolutionized efficiency via auxiliary-loss-free load balancing and multi-token prediction, further reducing training and inference costs. DeepSeek-R1 utilized GRPO technique, cutting GPU requirements to one-tenth of DeepSeek-V3. Each version demonstrates measurable gains of energy efficiency optimization in LLM development.

**Table 7**
LLM-focused software efficiency enhancement and their real-world applications.

| Software technique | Efficiency enhancement measure examples | Real-world LLM applications |
|---|---|---|
| Architecture | • MoE, e.g., Sparse MoE, load balancing<br>• Efficient attention, e.g., multi-head latent attention (MLA), hardware-assisted attention<br>• Long context LLMs, e.g., segmentation and sliding window | • Mixtral 8 × 7B<br>• DeepSeek-V2<br>• Grok-1<br>• Gemini 1.5 Pro |
| Pre-training | • Mixed precision acceleration<br>• Multi-token prediction<br>• Optimizers, e.g., Adam | • GPT-3<br>• OPT<br>• BLOOM |
| Model compression | • Quantization<br>• Parameter pruning<br>• Low-rank approximation<br>• Knowledge distillation | • LLaMA<br>• Qwen2.5<br>• QuantGPT |
| Parallelism | • Model parallel<br>• Pipeline parallel<br>• Data parallel | • Falcon<br>• Megatron-LM |
| Fine-tuning | • Parameter-efficient fine-tuning, e.g., low-rank adaptation<br>• Task-specific fine-tuning<br>• Supervised fine-tuning with proximal policy optimization or group relative policy optimization (GRPO) | • GPT-4o mini<br>• Mistral<br>• Claude 3 Haiku |

**Table 8**
Major software efficiency enhancement measures and effectiveness for DeepSeek series.

| Version | Launch date | Measures enhanced upon predecessor | Effectiveness |
|---|---|---|---|
| DeepSeek-V1 | January 2024 | Similar to LLaMA 2 | / |
| DeepSeek-V2 | May 2024 | MoE and MLA architectures | 42.5 % of training cost compared to V1 |
| DeepSeek-V3 | December 2024 | Auxiliary-loss-free strategy for load balancing and multi-token prediction | Training and inference costs about 1/10 of comparable models |
| DeepSeek-R1 | January 2025 | GRPO | 2000 GPUs required, which is 1/10 of V3 |

#### 4.1.2. Customized accelerator innovations

Accelerators exhibit remarkable efficiency in executing parallel processing tasks. They consume a substantial portion of the electricity utilized by LLMs, prompting extensive research into selecting optimal accelerators.

Selecting appropriate accelerators for different LLMs has garnered extensive research. Many practices have shown that GPUs significantly diminishes power consumption compared to CPUs [21,116]. Even among GPUs, while the H100 consumes more power in data centers than its predecessor, the V100, its productivity gains far outweigh this increase in energy usage. Furthermore, Google's custom TPUs exhibit an average energy efficiency that is 30–80 times higher than contemporary CPUs or GPUs [117].

Chip manufacturers' relentless pursuit of efficiency continues, as they introduce new accelerators tailored specifically for LLM applications, reducing the electrical energy required for training and inference of comparable-sized models [24]. For instance, Nvidia's release of the Blackwell GPU in March 2024, a state-of-the-art structure, not only effectively scales parameters of LLMs into the trillion-level range but also boosts performance for LLM inference workloads while slashing costs and power demand. Compared to an Nvidia chip released eight years ago, Blackwell may achieve a 350-fold reduction in energy used for training and an 45,000-fold decrease in energy used for inference [70,118].

The advancement of accelerators will contribute to mitigating the electricity growth associated with LLMs. However, when deciding on the optimal time to replace existing systems with more efficient alternatives, it is crucial to consider the entire lifecycle of hardware, necessitating careful evaluation and trade-off analysis [21]. For instance, by adopting a generalized formula for carbon emission rates specific to software systems like [119], the manufacturing electricity consumed by CPUs and accelerators throughout their entire lifecycles can be included.

Table 9 highlights hardware efficiency enhancements focused on LLMs and their real-world applications [120–123]. These improvements play a crucial role in enhancing the training and inference efficiency of LLMs, as well as altering their deployment methods, e.g., cloud-to-edge deployment which will be discussed in Subsection 4.2.2. Improved efficiency reduces energy consumption, while more flexible deployment options allow for better energy management of LLMs.

### 4.2. Optimization of physical architectures

#### 4.2.1. Power supply architecture transformation

Addressing the escalating power density challenges in LLM-specific data centers, enhancing the power supply architecture has emerged as a viable technical solution. For instance, directly employing 415/240V power transformers from medium-voltage distribution networks (e.g., 13 kV) may effectively meet the power demands of AI clusters. In North America, where hyper-scale data centers abound, adopting PDMs with higher voltage levels may represent a well-established strategy. This architectural refinement also mitigates power losses, enhancing the power efficiencies, thereby reducing electricity consumption.

Direct current (DC) power supply offers an alternative approach [124]. In contrast to traditional power transformers reliant on alternating current (AC), solid-state transformers, which are essentially power electronic converters, are compact in size and capable of AC-DC voltage conversion [125]. DC distribution promises an efficiency gain of 10 %–30 % over AC distribution. Given that data centers often integrate UPSs and renewable sources, this configuration better accommodates the proliferation of DC-based batteries and local photovoltaics, enhancing overall energy utilization.

Various energy efficiency enhancement initiatives contribute to

**Table 9**
LLM-focused hardware efficiency enhancement and their real-world applications.

| Hardware type | Efficiency enhancement measure examples | Real-world applications |
|---|---|---|
| GPU | • Tensor cores for mixed-precision matrix operations<br>• Tensor parallelism and pipeline parallelism<br>• Mixed-precision training adaption | • NVIDIA's A100 GPU |
| NPU | • Systolic array architecture for matrix multiplication<br>• Specialized hardware for Transformer layers<br>•Integration of quantization and pruning techniques | • Google's TPU v4<br>• Huawei's Ascend AI processors<br>• Apple's Neural Engine |
| FPGA | • Sparse matrix multiplication and quantization<br>• Customized for specific LLM model architectures<br>• Optimization of memory access patterns | • Microsoft's Project Brainwave |
| Memory | • Processing-in-memory and processing-near-memory<br>• Integration of computational logic into memory controllers<br>• Optimization of memory bandwidth and capacity | • Samsung's Aquabolt-XL, AXDIMM |
| Edge chips | • Design of specialized LLM accelerators<br>• Integration of NPUs<br>• Optimization of low-power LLM inference architectures | • Google's CXL-PNM Gemini Nano<br>• Apple's OpenELM<br>• Honor's MagicLM |

alleviating overall power demands as well. Prior to the rise of LLMs, enhancing energy efficiency of data centers, particularly their PUEs, was already a prime concern. As LLMs evolve, these efficiency-saving technologies remain crucial, facilitating the transition from small-scale, less efficient setups to hyper-scale data centers [10,126]. State-of-the-art solutions, e.g., hypervisor virtualization, advanced cooling techniques by algorithms such as deep Q-network [127], contribute to refining energy usage and mitigating power density requirements.

#### 4.2.2. Edge computing and cloud-to-edge deployment

Cloud computing has emerged as a crucial enabler for training LLMs [33]. Yet the scalability of data centers is inherently constrained, necessitating a focus on optimizing deployment architectures and integrating with technologies such as edge computing. Cloud-to-edge deployment alleviate LLM electricity demand by reducing data transmission latency and offloading computational tasks to distributed edge nodes, and thus minimizing reliance on energy-intensive central cloud data centers.

During the training stage, given its extremely intensive power demand of LLMs, ongoing developments like GPT-6 are exploring distributed training across distributed systems spanning thousands of accelerators and different clusters or data centers [128]. The training of LLaMA 3's 405 billion parameter model utilized distributed file systems and distributed prompts, with the computation process is also distributed [46]. This distributed deployment necessitates overcoming challenges like latencies, which has been addressed through techniques such as NVIDIA's InfiniBand adaptive routing and converged ethernet [45,129].

Shifting inference tasks from centralized AI clusters to edge devices represents a pivotal shift in deployment strategies. The cloud-to-edge architecture alleviates the pressure on hyper-scale data centers by delegating computational demands. Compared to conventional cloud-based LLM deployment, the proposed CE-CoLLM in Ref. [130] reduces inference time by 13.81 % and offloads over 84.53 % of computational tasks from cloud to edge. Case studies of the Edge-LLM in Ref. [131] shows a 17-fold improvement in overall computational speed, a 63 % reduction in the number of tasks experiencing timeouts, and up to a 43 % decrease in GPU overhead. Evaluations on the EdgeShard case in Ref. [132] give examples that it achieves 50 % lower latency and a twofold increase in throughput compared to baseline. This also aligns with the broader AI vision proposed in Ref. [133], which emphasizes leveraging local data and communication, thereby fostering scalable and resilient AI systems. According to Ref. [29], the workload at the edge may surge from the current 5 %–50 % by 2028.

While traditional deep reinforcement learning methods have facilitated offloading LLM inference to local edge devices, these solutions grapple with issues such as data inefficiency, insensitivity to latency requirements, and inflexibility in responding to workload fluctuation [134]. To address these issues, an algorithm resolving LLM inference offloading and resource allocation across cloud-edge systems is proposed in Ref. [135]. Reference [136] devises a distributed inference mechanism that harmonizes distributed computational capabilities with edge devices. Multilevel inference engines in Ref. [137] are integrated into edges to minimize resource demands.

Moreover, deployments of lightweight versions of developed LLMs or small language models (SLMs) [138], such as Microsoft's Phi-2 and Google's Gemini Nano, have commenced at the edge, enabling mobile devices like smartphones to execute functions previously exclusive to large models [139]. Another notable example is the Falcon model with 7 billon parameters, which can be operated on consumer-level devices like Apple's M2 [48].

Cloud-to-edge deployment not only reduces electricity demand but also enhances scalability, which is an essential advantage as the adoption of LLMs expands into edge environments such as the Internet of Things and mobile devices [140]. By prioritizing local processing of lightweight models like SLMs, such structure reliefs substantial energy pressure compared to cloud-only architectures, with minimal impact on accuracy. Specifically, the energy demand can be optimized significantly from two aspects, i.e., it alleviates the power burden on hyper-scale data centers in the cloud, while simultaneously improving the utilization of local renewable energy sources at the edge.

Nevertheless, concerns persist regarding the computational efficiency and energy efficiency of performing inference on resource-constrained edge terminals [141]. Training LLMs on energy-inefficient networks may exacerbate carbon emissions. Nonetheless, this deployment significantly mitigates the need for constructing or upgrading hyper-scale data centers, thereby alleviating pressure on local power distribution networks. Provided the technical advancements ensure the performance of LLMs, the energy efficiency trade-offs facilitated by the cloud-to-edge architecture merit further exploration.

### 4.3. Application of low-carbon energy accessibility

#### 4.3.1. Cleaner and closer power sources with lower carbon intensity

Clean power includes RESs such as wind turbines and photovoltaics, as well as large-scale power generators like nuclear energy units and high-efficiency gas turbines. Ensuring sufficient and relatively clean energy supply is crucial for guaranteeing sustainable green electricity supply.

Traditional approaches to sustainable electricity involve harnessing RESs. Leading companies, including Alphabet, Amazon, Meta, and Microsoft, have publicly committed to net-zero targets by purchasing carbon offsets to erase their historical carbon footprints [142–145]. For instance, Microsoft has committed to matching 100 % carbon-free electricity (CFE) purchase by 2030. In 2024, Microsoft agreed to support 10.5 GW of new renewable energy to support the sustainability of energy-intensive AI models [146].

But these carbon offsets based on CFE purchases are not enough to totally emit the environment impact of LLMs. Given concerns over power transmission congestion and losses in transmission and distribution, newly constructed hyper-scale data centers are being situated in regions with high proportions of renewable energy generation [19]. In China, a macro policy called "East Data, West Computing" has been strategically relocating data centers to renewable-rich western regions, leveraging abundant renewable energy resources [147]. Another notable case raises in the Arctic, where several data centers have already been built, exploiting wind energy and natural cooling conditions [148]. Such practical examples show the potential of combining renewable energy with energy-intensive infrastructure to address LLMs' long-term sustainability challenges.

Advancements in nuclear fusion technology or other low-carbon power sources may potentially decouple the development of LLMs from constraints imposed by sustained power supply. The stable load profile during LLM training aligns well with the characteristics of nuclear power generation. for instance, BLOOM utilized a data center in France powered by nuclear energy, thereby reducing its carbon footprint [149]. In the industry, case studies from AWS and Azure show that transitioning to 100 % renewable energy not only addresses short-term grid stress but also future-proofs LLM deployments against carbon pricing and regulatory shifts. For example, AWS's 960 MW nuclear-powered data center in Pennsylvania, the U.S., ensures stable and low-carbon power for LLMs over 30 years, eliminating long-term reliance on fossil fuels [150].

While gas turbines do not constitute absolute zero-carbon solutions, their adoption as large, high-efficiency units can contribute to mitigating carbon emissions for LLM power supply, particularly in addressing the immense electricity demands of hyper-scale data centers [151]. Strategies include selecting the least carbon-intensive energy grids [152], optimizing carbon emission through the geographic location of cloud service providers [15,32], and transparently disclosing power demand to end users when selecting LLMs.

Another approach involves co-locating data center construction with

local energy systems and using local power sources for direct electricity supply. In Ref. [153], when designing local integrated energy systems tailored to the cooling and power loads of a data center, it enhances both economic efficiency and reliability of energy supply. One recent case is that Google's facilities exemplify how strategic renewable energy adoption reduces LLM carbon footprints. In 2025, a total capacity of 724 MW solar power begins construction only one mile from Google's data center in Oklahoma, the U.S., which will be directly inject into the grid where it's consumed [154]. This is part of Google's carbon-free energy by 2030 initiative, which matches every hour of data center operations with renewable energy sources [155].

Meeting the urgent and intensive energy demand from rapidly expanding data centers requires establishing adequate and sustainable electricity supplies. Table 10 summarizes key power sources and their characteristics for data centers, including renewables. Currently, utility-scale solar or wind power and natural gas turbines offer the shortest construction timelines, aligning with typical data center construction schedules. While remotely-located large-scale renewable energy farms present environmentally sustainable options, transmission congestion from the power grid remains a constraint. Hydropower and nuclear plants typically entail decade-long development cycles, which are alternative options from a long-term environmentally-friend perspective. However, coal-fired generations, being the most carbon-intensive option, proves environmentally suboptimal for powering LLMs.

#### 4.3.2. *Economical and low-carbon energy management optimizations*

Pursuing cleaner and closer power sources necessitates a spatial approach, yet merely purchasing renewable electricity does not guarantee a data center's sustained and exclusive reliance on it [156]. A broader strategic objective for adopting low-carbon electricity necessitates temporal energy management, including shifting flexible workloads to time periods of higher renewable energy generation [157].

In the context of LLM's training, given their limited load flexibility, strategies involve adjusting training schedules, migrating loads across redundant zones, and *etc*. Seasonal fluctuations exist in the PUEs of data centers, with higher efficiency in winter than summer. For instance, the PUE of the data center used in Ref. [105] is 1.05 in January and 1.49 in July, respectively. In Ref. [127], an optimization framework base on deep learning algorithm is proposed to use more efficient PUE.

During the off-peak period for regular loads, electricity prices are typically lower, facilitating both cost reduction and low-carbon energy utilization for LLMs. The training process demonstrates flexibility through controlled checkpoints, enabling load balancing and energy savings aligned with the power curves of the grid or local energy systems. This includes delayed scheduling, workload elasticity, and the selection of power sources with the lowest carbon intensity [158–160]. Frameworks like Zeus can be employed to minimize the electricity during training. A virtual machine consolidation strategy with minimal slack time task scheduling is introduced by Ref. [161], effectively reducing power usage. A temporal shifting approach in Ref. [162] schedules data processing, computation, and server operation during periods of lower electrical load in the grid or high renewable energy availability. In Ref. [163], elastic carbon-aware workload scheduling techniques enhance power prediction and usage for data centers.

For unpredictable and relatively uncontrollable inference tasks in LLMs, UPSs like batteries and flywheels within data centers, designed for redundancy and security, become instrumental. These energy storage systems accumulate renewable energy during peak generation for utilization during off-peak periods. It is worth noting that most current energy management solutions focus on traditional data centers or various AI techniques, with relatively limited research focusing on LLMs. As LLMs have the potential to reduce the peak-to-average ratio of power loads in hyper-scale data centers to near 1.0, more targeted research can be promising in this specific area.

More importantly, from a global optimization perspective, electricity consumption and carbon emissions of LLMs can be optimized simultaneously. Reference [164] proposes an electricity-carbon collaborative demand response approach to achieve greater reductions in operation costs for both data centers and power systems. Factors on electricity and carbon emission from the power grid are incorporated during the planning stage of data centers in Ref. [165]. Reference [166] summarizes the potentials and mainstream methods for spatial and temporal demand response of large data center loads. Spatio-temporal optimizing considering both the electricity and carbon emission characteristics of the power grid facilitates cost savings and emission reduction for LLMs.

### 4.4. *Enhancement of external assessment and supervision*

#### 4.4.1. *Assessment indicators considering electricity demand*

Incorporating electricity-related assessment indicators into the consideration of both training and inference stages is crucial for enhancing the energy efficiency of LLMs [167]. Such refined indicators will facilitate the adoption of more energy-efficient LLMs by users. As revealed by Ref. [21], high-carbon LLMs do not always offer superior quality compared to their low-carbon counterparts.

While accurate measurement methods and international or national standards for electricity consumption and carbon emissions already exist from the perspectives of the power grid and data centers, standardized assessment frameworks for measuring the energy use and carbon footprints of specific LLMs and their lifecycle stages remain under development [168,169]. For instance, electricity metering in power grids relies on core standards such as IEC 62052 and IEC 62056 to ensure data accuracy. Carbon emissions are measured using a multi-tiered framework that includes ISO 14064, the Greenhouse Gas Protocol, and ISO 14067, providing comprehensive coverage from organizational to product-level assessments. From the perspective of data centers, the measurement of PUE is standardized under the ISO/IEC 30134 series. As to the LLM software, although various tools, e.g., CodeCarbon, MLCO$_2$, LLMCarbon, and Carbontracker, have emerged in recent years, they often produce inconsistent results for the same LLM due to differences in assessment methodologies and indicators [22,170–172]. Learning from the standardization practices of the power grid sector, several key

**Table 10**
Characteristics of electricity sources and sites for data centers.

| Sustainability type | Power source | Average carbon intensity ($kgCO_2eq/kWh$) | Construction timeline (Year) | Scheduling flexibility | Key constraints for siting |
|---|---|---|---|---|---|
| Renewable | Utility-scale solar or wind power | 0 | 1 to 4 | Non-dispatchable | Resource dependency |
| Renewable | Large-scale solar or wind power | 0 | 3 to 7 | Non-dispatchable | Power grid congestion |
| Renewable | Hydropower | 0 | 5 to 15 | Semi-dispatchable | Geographical constraint |
| Renewable | Geothermal power | 0 | 3 to 8 | Dispatchable | Tectonic zone limitation |
| Low-carbon | Nuclear plants | 0 | 5 to 15 | Dispatchable | Safety regulation |
| Fossil-based | Natural gas turbines | 0.39 to 0.62 | 1 to 4 | Dispatchable | Both pipeline infrastructure and power grid congestions |
| Fossil-based | Coal turbines | 0.96 | 3 to 6 | Dispatchable | Rail/port accesses, power grid congestion |

considerations should guide future efforts specifically for LLMs.

The assessment indicators should also tackle the issue of unifying factors. For instance, it is still unclear whether parameters such as PUE of data centers and accelerator years used are standardized in these assessment [23,173]. Furthermore, considering the carbon emissions from manufacturing infrastructure, a lifecycle assessment would impact the calculation of LLMs' electricity consumption. Although ex-post carbon offset or waste heat recovery practices are not directly related to the actual power consumption, they do influence the final figures. Therefore, it is recommended in Ref. [37] that research institutions and government departments collaborate to establish benchmarks for assessing the electricity costs of LLM's training and inference.

Moreover, to address the estimation discrepancies identified in Subsection 2.4, standardized evaluation frameworks for LLM electricity demand assessment should be established. Collaborative efforts among LLM developers, data center operators, and the broader community can advance standardization, as Table 11 illustrates.

1. LLM developers, as primary designers, play a fundamental role in reducing energy estimation variability. Disclosing software parameters ensures transparency. Normalizing hardware energy use to a reference baseline for legacy hardware allows cross-model and cross-generational comparisons. Additionally, reporting training and fine-tuning details, instead of a final aggregated value, clarifies the computational intensity behind energy consumption, enabling evaluations.
2. Data center operators, who is responsible for hosting LLM training and inference sometimes, must standardize carbon accounting to address geographic and operational variability. Reporting local power grid carbon intensity ensures emissions are tied to actual energy sources. Disclosing the temporal energy mix, e.g., hourly renewable penetration alongside PUE provides a complete picture of infrastructure efficiency. This combination prevents misleading emissions estimates and highlights the impact of grid decarbonization on LLM footprints.
3. A broader community, including researchers, policymakers, and standards bodies like ISO and IEC, drives consistency through shared guidelines and resources. Specifying assessment boundaries, e.g., inclusion or exclusion of hardware manufacture or carbon offsets, *etc.*, eliminates ambiguity in what constitutes one LLM's energy demand. Maintaining an open repository for energy-related datasets for aforementioned indicators also facilitates collaborative validation and benchmarking.

By aligning responsibilities across these stakeholders, such a framework can foster reliable and comparable evaluations of LLM electricity demand.

Another emerging solution emphasizes a balance between accuracy and energy usage in developing new LLMs or deploying existing ones. As LLMs' parameters and datasets balloon, achieving incremental accuracy gains becomes increasingly challenging [174]. Developers may prioritize model optimization over marginal accuracy improvements due to diminishing returns on investment for larger models [13]. The model in Ref. [175] consumed 2.5 times more FLOPs for image classification than that in Ref. [176], yet only reduces the error rate by 1.6 %. In Ref. [177], doubling the training batch size led to a mere 1 % performance boost on a benchmark test. Researchers are now examining whether the environmental and societal costs justify such enhancements [10]. Study in Ref. [105] show that constraining BERT's power usage results in a trade-off, i.e., an 8.2 % increase in time for a 12.0 % reduction in electricity consumption. Proposal in Ref. [101] have emerged to treat model precision as a function of computational budget, equating electricity demand with accuracy as an essential assessment indicator.

**Table 11**
Framework for standardizing LLM electricity demand evaluation.

| Stakeholder | Core focus | Key measures |
|---|---|---|
| LLM developers | Align software configurations and hardware benchmarks | Disclose model details<br>Normalize hardware energy use<br>Report training and fine-tuning details |
| Data center operators | Unify carbon emission accounting | Report local power grid carbon intensity<br>Disclose temporal energy mix alongside PUE |
| Broader community | Promote standardized assessment boundaries and collaboration | Specify assessment boundaries<br>Maintain an open repository for LLMs' energy-related datasets |

#### 4.4.2. *Transparency regulations from industry and government*

Enhancing the transparency of electricity information will facilitate the assessment of the electricity sustainability of LLMs, and empower users to make informed decisions regarding their carbon emissions when selecting LLM services [178]. However, prominent LLM operators tend to avoid reporting such information [4]. This lack of transparency partly stems from the operators themselves, who maintain strict confidentiality over LLM costs. Observed by Ref. [92], industry engagement in research publications is low, with only 23 % involving industry partners. Moreover, the reason for this phenomenon extends beyond LLM operators, as most of them lease data centers that do not disclose detailed data about power usage and carbon emissions. To foster competition beyond model quality, it is recommended in Ref. [19] that research papers in this field explicitly include carbon emissions.

Furthermore, the disclosure of electricity information should not merely increase transparency but also enhance granularity [10]. As demonstrated in Section 2, evaluating the electricity usage and carbon emissions of individual LLMs involves formula-based calculations, yet a significant portion of data relies on rough estimations. While some studies, such as those on OPT with 175-billion parameters [43], GPT-3 [32], Noor [179], and Gopher [47], have begun reporting $CO_2$ emissions, these disclosures often present final results only, necessitating a detailed breakdown that includes power, PUE, carbon intensity, and whether full lifecycle emissions of hardware manufacturing are considered, *etc.*, to enable more valuable comparisons [18,23].

Recognizing the drawbacks of closed systems and opacity within the competitive LLM landscape, legislators are increasingly aware of the need for disclosure. Governments should mandate or incentivize voluntary disclosure of data around the electricity consumption of LLMs, thereby enhancing transparency and empowering consumers with informed choices [4,7,180]. The UNESCO's "Recommendation on the Ethics of AI" advocates that business enterprises should assess the direct and indirect environmental impact, including carbon footprints and energy consumption [181]. While the EU's "AI Act" does not explicitly mandate AI energy reporting, they emphasize the transparency and efficiency of data centers [182].

Policymakers should also consider unintended consequences of LLM regulations on energy usage. Premature control without a comprehensive understanding the dynamics of LLMs' electricity demand could hinder performance improvements and limit their potential. For instance, measures like the U.S.'s "Blueprint for an AI Bill of Rights" and China's "Measures for the Administration of Generative AI Services", aimed at minimizing bias, hate speech, and personal data disclosure [174,183], may inadvertently increase computational costs during inference, escalating electricity expenses [184,185]. Additionally, policymakers should evaluate the substitution effect and rebound effects of LLM applications on sustainable power systems, transcending mere estimates of LLMs' marginal electricity consumption [10,186].

## 5. Conclusions and outlook

### 5.1. Conclusions

The emergence of LLMs represents one of the most disruptive technological advancements in recent years [187]. Their escalating computational demands underscores an impending surge in electricity demand, posing unignorable electricity and environmental implications [188]. Consequently, the accessibility and sustainability of electricity for LLMs have emerged as pressing issues [189].

Accurately evaluating the electricity consumption of LLMs is vital for fostering the healthy development of this field. While training LLMs may be perceived as a one-time endeavor, their inference applications span a vast user base, accumulating substantial ongoing electricity expenses over time. When discussing the electricity demand of LLMs, it is imperative to address both the immediate strain on power infrastructure due to their rapid proliferation and the long-term impacts stemming from advancements in LLM software, hardware, and externalities.

The opacity surrounding training stage, the non-disclosure of data for inference applications, and the inconsistency in estimations hinder the precision of evaluating LLMs' electricity demand. Nevertheless, comparative analyses of prevalent LLMs indicate that models with higher electricity usage and carbon emissions do not necessarily yield superior inference performance. Echoing technological history, LLMs, akin to other emerging technologies [65], will evolve from a phase of quantitative growth towards a pursuit of precision and energy efficiency, prompting developers to prioritize electrical and environmental benefits.

The sustainability of electricity in LLMs extends beyond the sole responsibility of developers. Upstream and downstream stakeholders, including data centers, power grid operators, and hardware manufacturers, share a corresponding duty for improvement. Moreover, policymakers are increasingly recognizing the significance of transparency in guiding regulatory strategies across the AI industry. In the context of LLMs, electricity sustainability should be mandatory, encompassing publicly reported electricity consumption, preferential development of energy-efficient hardware and algorithms, and enhanced utilization of RESs [190].

### 5.2. Future research directions and outlook

Several promising future research directions merit exploration to deepen the energy sustainability of LLMs, spanning from technical, architectural, and political aspects.

The first key direction is scaling low-carbon energy solutions for LLMs. While industrial applications like data centers in Arctic and Microsoft's solar-powered AI clusters demonstrate feasibility, scaling solutions remains unresolved. Future critical research areas include: hyper-scale energy management to optimize power distribution across renewable generations and energy storage systems, grid infrastructure upgrades to support terawatt-scale LLM demands, and environmental impact quantification spanning renewable infrastructure for AI hardware.

The second priority involves deploying hybrid cloud-to-edge inference technologies. Integrating edge devices with cloud servers offers significant potential to optimize energy consumption. Adaptive workload partitioning can allocate inference tasks between edge devices and cloud data centers. For example, simple queries like text classification can be offloaded to consumer hardware or local servers while reserving cloud resources for complex operations such as multi-modal generation. Implementing such hybrid structure mitigates demands for hyper-scale data center expansion, thereby reducing the need for extensive grid infrastructure upgrades while enabling utilization of distributed energy resources where edge devices locate.

Thirdly, policy-driven sustainability frameworks are also essential for global LLM ecosystems. Current initiatives from UNESCO and the EU provide a cross-border foundation, but key research gaps persist. Developing unified indicators to quantify LLM energy consumption may provide one potential solution to address inconsistencies in how countries or developers measure and report carbon footprints. Investigating incentive mechanisms, e.g., carbon pricing, tax breaks, and renewable energy quotas, can prioritize energy efficiency as well. Analyzing regulatory impacts and energy constraints for LLM performance advancements can be effective. It is important to engage policymakers, industry, and civil society in co-creating guidelines for sustainable LLM development.

By compiling data on electricity demand, analyzing challenges to future energy efficiency, and devising solutions to enhance energy performance, operators, researchers, infrastructure providers in the power sector and data centers, along with policymakers, can equip themselves with necessary tools to devise informed policies for sustainable power supply and zero-carbon green electricity in the realm of LLMs. This, in turn, will facilitate the design and deployment of more environmentally responsible LLMs, advancing the field towards a greener future.

## CRediT authorship contribution statement

**Zhenya Ji:** Conceptualization, Methodology, Writing – original draft, Project administration. **Ming Jiang:** Data curation, Visualization, Investigation, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

This work was financially supported by the National Natural Science Foundation of China under Grant 52107100 and 52077035, the Natural Science Foundation of Jiangsu Province of China under Grant BK20190710, and the Key Research and Development Program of Jiangsu Province under Grant BE2020081-4.

## Appendix

**Table A1**
Measured or estimated data for electricity consumption of training Falcon models, Mixtral 8 × 7B, and DeepSeek-V3

| Model | Falcon-40B | Falcon-180B | Mixtral 8 × 7B | DeepSeek-V3 |
|---|---|---|---|---|
| Accelerator type | GPU A100 | GPU A100 | GPU A100 | GPU H800 |
| Total number of accelerators | 384 | 4096 | 512 | 2048 |
| Training duration | 2 months | / | / | / |
| Efficacy | *70 %* | / | / | / |
| Accelerator hours (hour) | *387,072* | 7,000,000 | 552,960 | 2,788,000 |
| Accelerator TDP (W) | 400 | 400 | 400 | 350 |
| Electrical energy consumption excluding PUE (MWh) | **155** | **2800** | **221** | **976** |
| PUE | *1.05* | *1.05* | *1.05* | *1.05* |
| Electrical energy consumption including PUE (MWh) | **163** | **2940** | **232** | **1025** |
| Carbon intensity ($kgCO_2eq/kWh$) | 0.05 to 0.55 | 0.05 to 0.55 | 0.027 | 0.5366 |
| Carbon emission (ton) | **8 to 90** | **147 to 1617** | **6.3** | **550** |

Notes.
1. Roman type indicates values derived from public literature as indexed in Table 3.
2. Italicized type indicates values that are not publicly disclosed and are estimated based on relevant technical pathways.
3. Bold type represents estimations calculated using the formula for online measurement approach in Subsection 2.2.2. That is, multiplying Accelerator hours by Accelerator TDP yields the Electrical energy consumption excluding PUE. Taking Mixtral 8 × 7B as an example: 552,960 h × 400 W ÷ 1000 = 221,184,000 Wh, resulting in approximately 221 MWh.
4. The symbol "/" indicates that the accelerator hours already have data publicly disclosed by developers or estimated in other literature, eliminating the need of using this sub-item. In specific, for Falcon-40B which involves this calculation, 384 GPUs × 2 months × 30 days/month × 24 h/day × 70 % = 387,072 GPU hours. The reason for multiplying by the efficacy is that GPU hours are typically normalized to full load.
5. For PUE, an industry-leading value is adopted.
6. The Falcon series were trained on AWS SageMaker, though the geographic location remains unspecified; thus, a carbon intensity range of 0.05–0.55 $kgCO_2eq/kWh$ in AWS's global infrastructure was adopted to account for spatial variability. The significant disparity in carbon intensity between France and China stems from substantial differences in their energy mixes. France relies heavily on nuclear power, whereas China has a much higher proportion of coal-fired power.
7. The value of "Electrical energy consumption including PUE" is used in Table 3, as this metric directly correlates with the total carbon emissions.

## Data availability

Data will be made available on request.

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